

Liquid Chromatography-Mass Spectrometry for Bacteria Recognition

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Abstract. Rapid identification of urinary pathogens from LC-MS data could shorten diagnosis time compared with standard culture methods. However, automatic classification is difficult due to high-dimensional signals, strong batch effects, and class imbalance.

We study these challenges on LC-MS urine data collected across 13 batches and 28 bacterial classes, combining artificial samples with a limited set of real patient samples. We compare several deep learning approaches, including PCA, autoencoders, and encoder-based classifiers. We find that reconstruction-based latent spaces are not sufficient for accurate classification. In contrast, a simple batch correction—subtracting the mean blank signal per batch—combined with compact bottleneck classifiers trained for discrimination yields the most stable performance. The best model reaches 83.6% accuracy and 83.7 macro-F1 on the artificial samples. When applied to real patient subset, the model achieves about 60% accuracy, showing that performance remains limited on clinical samples.

Keywords: LC-MS · Deep Learning · Bacteria Detection · Dimensionality Reduction · Batch Effects · Autoencoder

1 Introduction

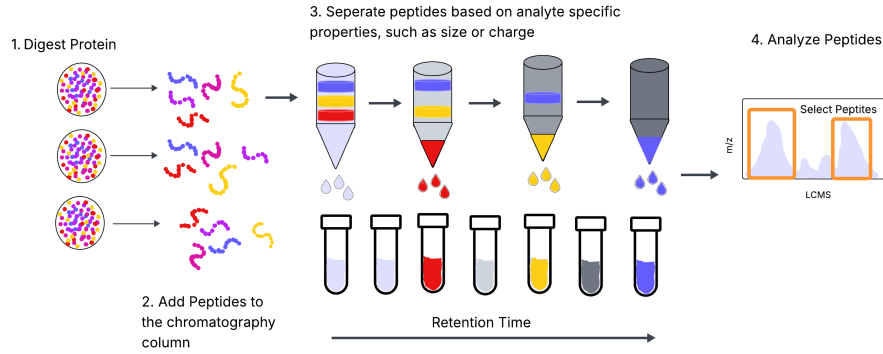
Liquid chromatography-mass spectrometry (LCMS) is a fundamental analytical technique that integrates the separation power of liquid chromatography with the molecular specificity of mass spectrometry. [13] In this process, a biological sample is injected into a high-pressure liquid stream (called the mobile phase) which passes through a chromatographic column filled with a solid material (called the stationary phase). Different compounds (e.g. peptides, proteins,...) in the sample interact with this material in different ways and at different strengths, causing them to move through the column at different speeds and become separated over time. [13,7]

Once separated, these compounds leave the column one by one and are converted into charged particles before entering a mass spectrometer (see Figure 1a). The mass spectrometer measures the mass-to-charge ratio (m/z) of these charged particles and produces intensity spectra that provide detailed information about

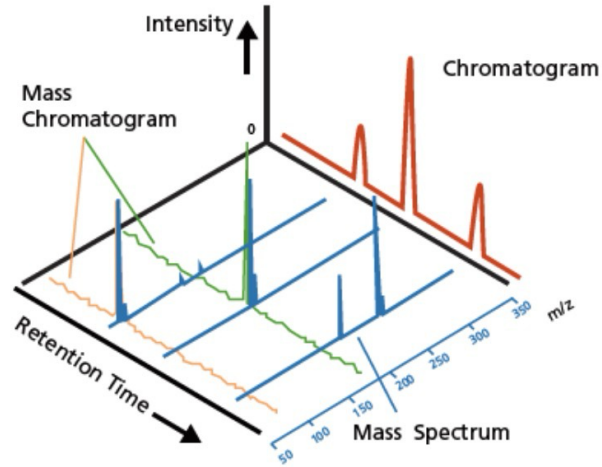
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the molecular makeup of the sample. Other mass spectrometry techniques such as matrix-assisted laser desorption/ionization time-of-flight (MALDI-TOF) are also widely used. [7]

LCMS data is typically represented as a series of intensity histograms over retention time (RT) and m/z axes (see Figure 1b). [13,7] These histograms capture the intensity of detected peptides. However, they are not built at the precision of each individual peptide. Instead, peptides are aggregated based on their m/z values, meaning that a single peak in the histogram may correspond to several different peptides with similar m/z values.



(a) LCMS detailed workflow.



(b) Typical LCMS data representation [15].

This study focuses on urine samples collected from pregnant women to detect urinary tract infections (UTIs) caused by bacterial pathogens. However, the application of LCMS for bacterial species identification in these clinical samples presents three major challenges. First, the high dimensionality of LCMS data is substantial: each sample generates approximately 100,000 features creating a significant "curse of dimensionality" for machine learning models [2]. Second, batch effects arising from differences in acquisition conditions such as measurements taken on different days, using different instruments, or by different operators introduce systematic biases into the data. These batch effects add non-biological variability that confuses models and reduces the reproducibility and reliability of results, making it hard to separate true biological differences from technical noise [4] [12]. Third, the multiclass nature of the problem with 28 possible bacterial species. The objective of this study is to develop and evaluate deep learning methods that can effectively predict bacterial species present in urine samples from LC-MS data, while competently addressing the challenges of high dimensionality and batch effects.

2 Related Works

2.1 Handling High Dimensionality in Mass Spectrometry Data

The most used approach for dimensionality reduction in mass spectrometry-based studies is Principal component analysis (PCA) though other methods, such as singular value decomposition and non-negative matrix factorization, have been used to a great degree as well. [1]. Gómez-Caudillo et al. [3] applied PCA to LCMS data from glioblastoma cell lines and compared it with 2D electrophoresis coupled to MALDI-TOF. Surprisingly, their analysis showed that PCA could yield information of comparable quality to the more labor-intensive proteomic techniques, suggesting that linear dimensionality reduction methods can effectively capture essential biological variation in mass spectrometry datasets. However, PCA assumes linear relationships among features, which may not always hold for complex biological mass spectra.

Deep learning approaches, particularly autoencoders and their variants, offer more flexible dimensionality reduction by learning non-linear mappings from high-dimensional data to compact representations. The denoising autoencoder (DAE) architecture specifically addresses the noise inherent in mass spectrometry data by training the model to reconstruct clean data from corrupted inputs, thereby learning robust feature representations [8]. Variational autoencoders (VAE) extend this approach by learning probabilistic representations of the data, providing additional benefits for generative modeling and uncertainty quantification. These neural network-based methods can simultaneously reduce dimensionality while extracting features that are optimized for downstream classification tasks [1].

In our specific challenge of 100,000 dimensional LCMS features, learned encoders present advantages over manual feature selection because they can discover complex and non-obvious feature combinations. However, the choice be-

tween linear (PCA) and non-linear (autoencoder-based) methods remains dataset dependent and should therefore be validated empirically.

2.2 Machine Learning and Deep Learning for Mass Spectrometry Classification

Recent years have witnessed a dramatic expansion in the application of machine learning and deep learning methods to mass spectrometry-based classification tasks. Beck et al. [1] provide a comprehensive overview of machine learning methodology relevant to mass spectrometry, covering both classical approaches and modern deep learning architectures. Their review highlights how traditional machine learning methods including support vector machines (SVM), random forests, and ensemble classifiers continue to serve as baselines, while new neural network architectures such as CNN, LSTM and Auto-Encoder have enabled more sophisticated feature learning directly from raw or minimally pre-processed data.

Seddiki et al. [14] developed a combined CNN-LSTM architecture for mass spectrometry based tissue classification that addresses key challenges including high dimensionality, peak shifts between acquisitions, and noise. The CNN component significantly reduces dimensionality while learning spatially relevant features, while the LSTM network captures temporal patterns in the data. This architecture achieved strong results without requiring a preprocessing for feature learning. Similarly, Li et al. [8] achieved 100% classification accuracy for *Listeria* species identification by incorporating DAE with SVM applied to MALDI-TOF MS data.

At the preprocessing level, Melnikov et al. [10] developed "peakonly" a CNN approach for precise peak detection in high-resolution LC-MS data. Their method achieved 97% accuracy in distinguishing true positive peaks from false noise peaks.

2.3 Batch Effects in LCMS Analysis

Batch effects or systematic non-biological variations arising from differences in sample processing protocols, acquisition conditions, instrument drift, and operator variations represent a critical challenge in large scale mass spectrometry studies. These artifacts can overwhelm true biological signals, compromising reproducibility and confounding statistical inference. Han and Li [4] provide a comprehensive review of batch effect origins, detection approaches, and correction methods in metabolomics. Their analysis distinguishes between inter-batch effects (differences between separate analytical runs) and intra-batch effects (systematic drift within a single acquisition session), each requiring tailored correction strategies.

Traditional batch effect correction methods include normalization approaches, transformation procedures, and quality control (QC) based strategies. Kuligowski et al. [6] developed QC-SVRC (quality control samples with support vector regression for correction), which uses repeated analysis of QC samples to model and

correct intra-batch instrumental drift. By fitting a regression model to known QC samples distributed throughout an analytical batch, this approach can retrospectively correct systematic intensity changes while preserving biological variation. Similarly, Mertens [11] provided a critical review of transformation, normalization, and batch correction procedures, emphasizing that the choice of method significantly impacts downstream analysis and that no single approach is universally optimal.

More recently, in 2024 neural network based approaches have emerged as powerful tools for batch effect removal. Pelletier et al. [12] introduced Batch Effect Removal Neural Networks (BERNN), a suite of deep learning models specifically designed to remove batch effects in large LCMS experiments while preserving genuine biological variation and maximizing sample classification performance. Their comprehensive comparison across five diverse datasets demonstrated that BERNN models consistently achieved the strongest sample classification performance compared to existing batch correction methods. Importantly, Pelletier et al. found that the model producing the greatest batch effect removal did not always yield the best downstream classification. Tsai et al. [16] emphasize that proper handling of batch effects must begin during the data acquisition and preprocessing phases, not merely as a post-hoc correction.

3 Data and Preprocessing

3.1 Dataset Description

The dataset consists of urine samples collected from pregnant women undergoing screening for urinary tract infections. Raw LC-MS measurements are stored as tab-separated files of $(rt, m/z, \text{intensity})$ triplets, one file per sample. Samples are distributed across **13 acquisition batches**, each batch being acquired on a potentially different day, by a different operator, or on a different instrument. The overall dataset covers **28 bacterial species** with highly uneven class distribution. Each batch also contains a set of *blanc* samples. These are not bacteria samples, but reference acquisitions performed without biological material. Their role is to capture the technical background of a given acquisition batch, including instrument noise, solvent-related signal, and batch-specific drift. As shown later in the paper, these blanc samples become central for handling batch effects.

3.2 Binning and Feature Extraction

Raw $(rt, m/z, \text{intensity})$ triplets are converted into fixed-length feature vectors by discretisation:

1. the RT and m/z ranges are divided into bins of widths Δrt and $\Delta m/z$;
2. intensities inside each (rt_i, mz_j) cell are summed;
3. the resulting intensity map is flattened into one sparse vector;
4. all batches are merged into one machine-learning matrix.

We explored three resolutions, all with `mz_bin = 10` and different retention-time bin widths. After removing all-zero columns, the final feature counts are given in Table 1.

Table 1: Dataset variants and their feature dimensionalities.

Dataset	Original features	After zero-column removal
mzp2 (<code>rt_bin = 2</code>)	851 620	386 962
mzp10 (<code>rt_bin = 10</code>)	171 500	87 743
mzp320 (<code>rt_bin = 320</code>)	14 700	6 249

4 Methodology

Our methodology evolved progressively rather than through a single architecture search. Each stage was designed to test a more precise hypothesis about what limited performance on this LC-MS classification problem: dimensionality reduction, latent supervision, class imbalance, batch effects, or the interaction between them.

4.1 Two-Step Pipelines

The first family of models follows a classical two-step paradigm in which dimensionality reduction and classification are optimized separately. We first reduce the input using either PCA or a standalone autoencoder, then train a multilayer perceptron (MLP) on the resulting representation.

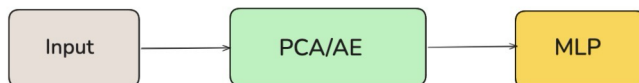


Fig. 2: Two-step pipeline: independent reduction followed by MLP classification.

This stage serves as both a baseline and a first test of whether an unsupervised latent space learned only through reconstruction can preserve the information needed for bacterial discrimination.

4.2 Joint Encoder-Decoder Classifiers

The limitations of the two-step autoencoder baseline motivated a second family of models in which representation learning and classification are trained jointly.

We merged encoder, decoder, and classifier into one end-to-end model:

$$\mathcal{L} = \mathcal{L}_{\text{recon}} + \lambda_{\text{cls}} \mathcal{L}_{\text{CE}}. \quad (1)$$

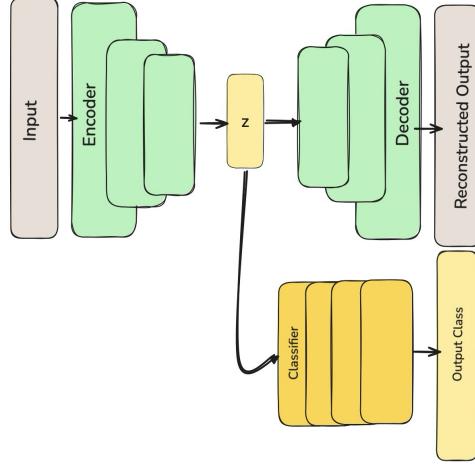


Fig. 3: Joint encoder-decoder-classifier architecture used in the early end-to-end stage.

In this setting, classification gradients also shape the encoder, forcing the latent representation to keep class-relevant information rather than optimizing only for reconstruction fidelity.

4.3 Loss Engineering and Additional Regularization

Because the classification problem is strongly imbalanced, we next investigated loss functions designed to improve minority-class discrimination and to structure the latent space more explicitly.

Focal Loss. To counter class imbalance, we replaced standard cross-entropy with Focal loss [9], which introduces a modulating factor $(1 - \hat{p}_t)^\gamma$ that down-weights samples already classified correctly and forces the model to focus more strongly on difficult or rare examples:

$$\mathcal{L}_{\text{focal}} = -\alpha_t (1 - \hat{p}_t)^\gamma \log \hat{p}_t. \quad (2)$$

Here, $\hat{p}_t$ is the predicted probability of the true class, α_t is a class-balancing factor, and γ controls how strongly easy samples are down-weighted.

Supervised Contrastive Loss. We also evaluated supervised contrastive loss (SupCon) [5] in order to organize the latent space more explicitly. SupCon pulls together embeddings of samples belonging to the same class while pushing apart embeddings from different classes, thereby encouraging tighter and better separated class clusters:

$$\mathcal{L}_{\text{SupCon}} = \frac{-1}{|P(i)|} \sum_{p \in P(i)} \log \frac{\exp(\mathbf{z}_i \cdot \mathbf{z}_p / \tau)}{\sum_{a \in A(i)} \exp(\mathbf{z}_i \cdot \mathbf{z}_a / \tau)}. \quad (3)$$

In this expression, $P(i)$ denotes the set of same-class positive samples for anchor i , $A(i)$ is the set of all other samples in the mini-batch, and τ is a temperature hyperparameter.

Combined objective. These loss functions were integrated into the joint encoder-decoder setting by combining reconstruction and classification-oriented supervision:

$$\mathcal{L} = \mathcal{L}_{\text{recon}} + \lambda_{\text{cls}} \mathcal{L}_{\text{focal}} + \lambda_{\text{sup}} \mathcal{L}_{\text{SupCon}}. \quad (4)$$

Additional regularization. In addition, we explored more experimental regularizers, including gradient-reversal and alignment-inspired losses such as optimal-transport-like terms. These extensions were motivated by the idea that a better structured latent space might improve both minority-class recognition and robustness to distribution shift.

4.4 Variational and Conditioned Encoder-Decoder Models

We next investigated richer encoder-decoder families in order to test whether a more structured latent-variable formulation could improve classification under strong dimensionality constraints. In particular, we explored joint variational autoencoders (JVAE) and conditional variational autoencoders (CVAE).

Joint Variational Autoencoder (JVAE). The JVAE extends the previous joint encoder-decoder classifier by replacing the deterministic latent code with a probabilistic latent variable. Instead of producing a single embedding vector, the encoder outputs a mean vector μ and a log-variance vector $\log \sigma^2$. A latent representation is then sampled using the reparameterization trick:

$$z = \mu + \sigma \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I). \quad (5)$$

A Kullback-Leibler regularization term is added to encourage the latent space to remain close to a standard normal prior:

$$\mathcal{L}_{\text{KL}} = -\frac{1}{2} \sum_j (1 + \log \sigma_j^2 - \mu_j^2 - \sigma_j^2). \quad (6)$$

The objective retains the same combination of reconstruction, variational regularization, focal loss, and supervised contrastive supervision:

$$\mathcal{L} = \mathcal{L}_{\text{recon}} + \lambda_{\text{kl}} \mathcal{L}_{\text{KL}} + \lambda_{\text{cls}} \mathcal{L}_{\text{focal}} + \lambda_{\text{sup}} \mathcal{L}_{\text{SupCon}}. \quad (7)$$

Conditional Variational Autoencoder (CVAE). The CVAE goes one step further by explicitly conditioning the latent model on class information. In practice, a class embedding or label-conditioned vector is injected into the encoder and decoder, so that the latent representation is learned relative to the bacterial class. The underlying hypothesis was that conditioning could help disentangle class-specific variation from other sources of variation, including noise and batch-dependent effects.

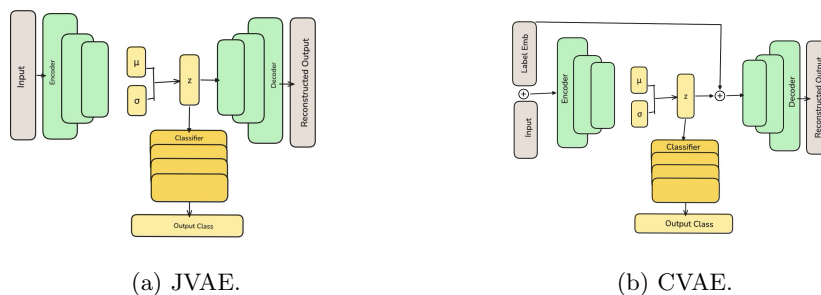


Fig. 4: Representative variational encoder-decoder architectures explored during the project.

These models were designed to test whether a more regularized and class-structured latent space could outperform simpler encoder-decoder architectures. They improved over the weakest baselines, but they did not become the dominant solution in the final benchmark.

4.5 Blanc-Aware Batch Effect Strategies

A major challenge of this dataset is the presence of strong batch effects. Because each batch contains several blanc reference samples, we investigated multiple ways to use them explicitly for batch correction. The underlying hypothesis was that the blanc captures a large part of the technical signature of a batch and can therefore help separate non-biological variation from the bacteria-related signal.

We explored four main strategies.

Direct subtraction. The simplest strategy consists in subtracting the mean blanc signal of the batch from each sample, yielding the corrected representation $x - \bar{b}$. This approach treats the blanc as an estimate of the additive technical background shared by samples acquired in the same batch.

Concatenated representation. Instead of replacing the original signal, we also tested a richer input of the form $[x, x - \bar{b}]$. The motivation was to preserve both the raw signal and its batch-corrected counterpart, allowing the classifier to exploit whichever combination is most informative.

Dual-encoder architectures. We next explored architectures in which the sample and the corresponding blanc are processed by separate encoders. Their latent representations are then combined before classification. The idea was to let the model learn explicitly how to use, discount, or separate batch-specific information rather than fixing the correction entirely at preprocessing time.

Multitask variants. Finally, we designed multitask models intended to promote batch-invariant representations through blanc-to-blanc prediction. The core idea was to use blanc signals from different batches and train the model to reconstruct the blanc of one batch from the blanc of another batch. Since these blanc samples do not contain biological signal and should represent the same type of reference acquisition, the model is encouraged to focus on their shared structure rather than on batch-specific nuisance variation. In parallel, the same model also performs bacteria classification. The underlying hypothesis was that this auxiliary task would force the latent space to retain more batch-invariant information and reduce sensitivity to technical drift.

Across these experiments, the most effective solution turned out to be the simplest one: direct preprocessing with the mean blanc subtraction $x - \bar{b}$. More elaborate architecture-level uses of blanc information were well motivated, but they did not provide more reliable gains than this simple correction.

4.6 Classifier-First Bottleneck Models

After the blanc-aware experiments, a recurring limitation remained the same: the latent space was still being shaped too strongly by reconstruction. Across multiple encoder-decoder models, we repeatedly observed that reducing the weight of the reconstruction loss improved classification performance. This suggested that the main difficulty was not simply model capacity or loss design, but the fact that reconstruction and classification were competing for control of the latent representation.

To investigate this more carefully, we performed a series of simple but decisive experiments in which the reconstruction objective was progressively weakened. In particular, we tuned the reconstruction weight over increasingly smaller values and consistently observed that lower weights produced better classification results. This led to two important follow-up experiments.

First, we tested a delayed-reconstruction regime in which the classification objective is optimized alone during the first training epochs, while the decoder loss is frozen. Reconstruction is only introduced later with a very small weight. The goal of this strategy was to allow the encoder to first learn a discriminative representation before being regularized by reconstruction.

Second, we considered the limiting case in which the reconstruction weight is set to zero. In practice, this amounts to removing the reconstruction objective entirely and keeping only a feature extractor followed by a classifier. Rather than degrading performance, this surprisingly produced the strongest and most stable family of models in the project.

The resulting architecture is a classifier-first bottleneck model trained on either $x - \bar{b}$ or $[x, x - \bar{b}]$. In the stable final family, the input is first projected to an intermediate hidden representation, compressed into a small latent bottleneck, and then classified directly with a lightweight head. The dominant supervision is weighted cross-entropy, with no reconstruction objective.

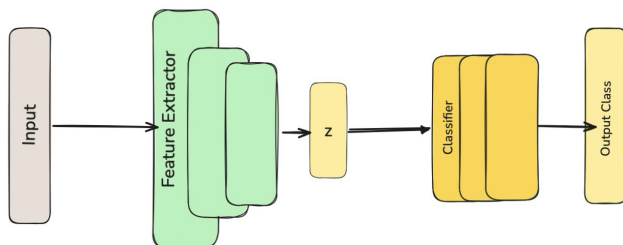


Fig. 5: Representative classifier-first bottleneck architecture used in the final phase.

5 Experiments

5.1 Experimental Setup

The experiments were conducted on three dataset resolutions derived from the same LC-MS acquisition pipeline: **mzp320**, **mzp10**, and **mzp2**. The performance was evaluated primarily with two complementary metrics: accuracy and macro-F1. Accuracy provides a global view of classification performance, while macro-F1 is particularly important in this setting because it gives equal weight to all bacterial classes despite strong class imbalance.

5.2 IID Benchmark Across Resolutions

The main benchmark protocol of the project is an IID evaluation setting in which train, validation, and test splits are drawn from the full dataset while keeping all acquisition batches represented across the splits. This protocol is used consistently to compare model families across the three dataset resolutions and therefore provides the main basis for the benchmark results.

The purpose of this benchmark is twofold. First, it measures how model performance evolves as the resolution increases from **mzp320** to **mzp2**. Second, it makes it possible to compare the different architectural families under a common setting: two-step reduction pipelines, joint encoder-decoder models, variational variants, blanc-aware models, and final classifier-first bottleneck networks.

5.3 Patient-Oriented Evaluation

In addition to the IID benchmark, we designed specific experiments centered on the real-patient samples contained in batch b10. This subset is particularly important because it allows us to test whether models trained mainly on artificial bacteria samples can generalize to real clinical samples.

We considered several patient-oriented protocols. In a first setting, patient samples are included in the usual IID splitting procedure, making it possible to evaluate patient performance when similar patient examples are present in training. In stricter settings, patient samples are held out from training and used only for final testing, while the model is trained either on all artificial bacteria data or on artificial data augmented with a very small number of patient samples. These protocols are intended to separate ordinary IID performance from true patient generalization.

6 Results

6.1 Representative Benchmark Across the Three Resolutions

Table 2 summarizes the main benchmark results obtained throughout the project under the standard IID protocol. It gathers the main model families we evaluated, together with several ablation variants, across the three dataset resolutions `mzp320`, `mzp10`, and `mzp2`.

Table 2: Full benchmark results: accuracy / macro-F1. Best result per dataset column is shown in bold.

Model	mzp320	mzp10	mzp2
PCA + MLP	0.575 / 0.572	0.551 / 0.548	0.559 / 0.554
AE + MLP	0.229 / 0.221	0.235 / 0.222	0.227 / 0.206
AE + MSE + CE	0.593 / 0.591	0.636 / 0.632	0.662 / 0.661
AE + Focal	0.611 / 0.606	0.634 / 0.635	0.621 / 0.620
AE + Focal + SupCon	0.597 / 0.588	0.627 / 0.617	0.605 / 0.598
JVAE	0.656 / 0.660	0.648 / 0.650	0.648 / 0.651
CVAE	0.652 / 0.656	0.696 / 0.704	0.684 / 0.688
JAE Dual encoder	0.627 / 0.630	0.658 / 0.662	0.650 / 0.654
JAE ($x - \bar{b}$)	0.627 / 0.631	0.713 / 0.715	0.723 / 0.730
JAE (no res-blocks)	0.595 / 0.607	0.623 / 0.630	0.615 / 0.616
JAE (Fusion concat)	0.642 / 0.644	0.666 / 0.664	0.650 / 0.652
JAE multitask	0.591 / 0.581	0.623 / 0.610	0.603 / 0.608
JAE two-stage	0.599 / 0.596	0.640 / 0.633	0.648 / 0.647

This first benchmark shows the overall progression of the project, from simple baselines to joint encoder-decoder models and later blanc-aware variants. We then focus on the final best-performing models obtained on the main high-resolution benchmark, `mzp2`.

6.2 Final Results on the Highest-Resolution Dataset

The results in Table 3 correspond to the best model obtained in this work.

Table 3: Final classifier-first encoder results on `mzp2`.

Preprocessing	Architecture	Accuracy	Macro-F1
$x - \bar{b}$	Bottleneck-64	0.822	0.825
$[x, x - \bar{b}]$	Bottleneck-64	0.820	0.818

6.3 Patient-Oriented Experiments

The results show that this setting is much harder than the standard IID case. When patient samples are fully held out, performance drops sharply, indicating a clear gap between artificial and real-patient data.

However, this gap is not fixed. In the few-shot setting, adding a small number of patient samples during training improves performance on the remaining held-out patients. In our experiments, this setup reached patient accuracies of up to about 60%.

7 Conclusion

We presented a progressive study of deep learning for LC-MS bacteria recognition in urine samples, spanning classical baselines, joint encoder-decoder models, variational variants, and final blanc-aware discriminative models. The main result of this study is that simple choices worked best: direct batch correction with $x - \bar{b}$ and classifier-first bottleneck models consistently outperformed more elaborate reconstruction-first designs.

On the highest-resolution dataset, the best IID result reaches 82% accuracy and 82% macro-F1, while the best classifier-first bottleneck models remain close behind and form the strongest stable family overall. At the same time, the patient-oriented experiments show that transfer to real clinical samples remains substantially harder.

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